TEXORA · LECTURE SHEET

Circular Knitting Machine Calculations

Production of a circular machine from first principles: courses per minute from rpm and feeders, yarn per course from needles and stitch length, and the two made consistent — kilograms per shift checked against metres, width and GSM, so an error in any one number exposes itself in the others.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 8 · SPECIALIZATION — FABRIC MANUFACTURING

1. The reference machine

PARAMETER	VALUE
Cylinder	30 in diameter, 24 gauge → needles = π × 30 × 24 = **2,262** (Sheet 7)
Feeders	90
Speed	20 rpm
Yarn	Cotton Ne 30 (19.68 tex), single jersey
Stitch length	2.9 mm
Efficiency	85%

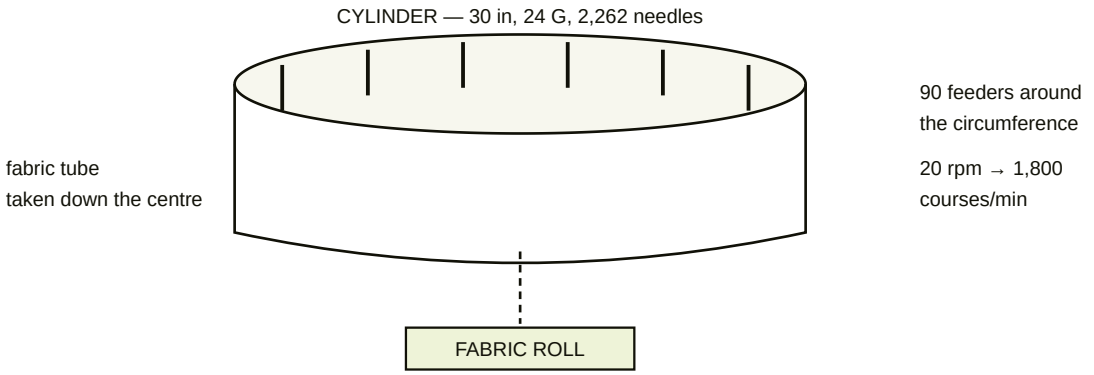


Figure 1 — The reference machine in section: cylinder and needles above, fabric drawn down to the roll.

2. The production chain

$$\text{Courses per minute} = \text{rpm} \times \text{feeders}$$

$$\text{Yarn per course (m)} = \text{needles} \times \text{stitch length (mm)} \div 1{,}000$$

$$\text{Production (kg/h)} = \text{courses/min} \times \text{yarn per course} \times 60 \times \text{tex} \div 10^6 \times \text{efficiency}$$

WORKED EXAMPLE — PRODUCTION OF THE REFERENCE MACHINE

Courses per minute = 20 × 90 = **1,800**.

Yarn per course = 2,262 × 2.9 ÷ 1,000 = **6.56 m**.

Yarn consumption = 1,800 × 6.56 = **11,808 m/min**.

At 100%: 11,808 × 60 × 19.68 ÷ 10^6 = **13.94 kg/h**.

At 85% efficiency: 13.94 × 0.85 = **11.85 kg/h** → per 8-hour shift = **94.8 kg**.

This machine and this shift figure run the whole knitting chain of the course: Sheet 10 builds an order on them, Sheet 12 plans a fabric around their method, Sheet 14 houses them in a shed.

3. The consistency check — metres, width, GSM

A knitting calculation that cannot cross-check is a guess. Close the loop with the structure relations of TE-3205: for cotton jersey at full relaxation, CPC × l ≈ 56, and GSM = WPC × CPC × l × tex ÷ 100.

WORKED EXAMPLE — CLOSING THE LOOP

CPC = 56 ÷ 2.9 = **19.3 per cm**.

Fabric length per hour = courses per hour ÷ courses per metre = (1,800 × 60) ÷ (19.3 × 100) = **55.9 m/h** (at 100%).

At WPC 12 per cm, open width = 2,262 ÷ 12 = 188.5 cm = **1.89 m**; GSM = 12 × 19.3 × 2.9 × 19.68 ÷ 100 = **132 g/m²**.

Independent production = 55.9 m/h × 1.89 m × 132 g/m² = **13.94 kg/h** — identical to Section 2, as it must be: the two routes count the same yarn, once by the course and once by the cloth.

Efficiency discipline. Note where the 85% sits: it multiplies the *rate* (kg/h, kg/shift) but never the *structure* (CPC, width, GSM at a given stitch length). A plan that lets efficiency leak into GSM will order the wrong yarn for the right fabric.

4. What changes the answer

CHANGE	EFFECT ON KG/H	EFFECT ON FABRIC
rpm 20 → 24	+20% (rate × 1.2)	None, if yarn feed per course is maintained
Feeders 90 → 72	−20%	None structurally; fewer courses per revolution
Stitch length 2.9 → 3.2 mm	+10% yarn per course	Looser fabric: CPC falls to 56/3.2 = 17.5, GSM falls (Sheet 9 takes this up)
Count Ne 30 → Ne 24 at same l	+25% (tex ratio)	Heavier, tighter fabric — K rises; a different cloth, not just a faster one

Glossary

Course — one row of loops from one feeder pass. **Feeder** — a yarn supply station; courses per revolution. **Yarn per course** — needles × stitch length. **Efficiency** — productive time share; multiplies rates, never structure. **Open width** — tube circumference expressed flat: needles ÷ WPC.

Key points

• Courses/min = rpm × feeders (1,800); yarn/course = needles × l (6.56 m).
• Production = courses × yarn/course × tex (13.94 kg/h at 100%; 94.8 kg per shift at 85%).
• Always close the loop: length × width × GSM must return the same kg/h (55.9 × 1.89 × 132 = 13.94).
• rpm and feeders change rate only; stitch length and count change the cloth itself.

Practice

1. The machine is re-set to 24 rpm at the same stitch length and efficiency. Find the new production per shift.
2. A 26-inch, 28-gauge machine (2,287 needles) runs 72 feeders at 18 rpm, stitch length 3.42 mm, Ne 24 (24.6 tex), 85% efficiency. Find its kg per shift. (Sheet 12 plans a whole fabric on exactly this machine — compare when you get there.)
3. Why must the Section 3 check agree exactly, and what does a disagreement of 10% tell the technologist?

Answer key

1. Rate scales with rpm: 94.8 × 24/20 = **113.8 kg per shift**.
2. Courses/min = 18 × 72 = 1,296; yarn/course = 2,287 × 3.42 ÷ 1,000 = 7.82 m; consumption = 1,296 × 7.82 = 10,135 m/min; kg/h at 100% = 10,135 × 60 × 24.6 ÷ 10^6 = 14.96; at 85% = 12.72 kg/h →

101.7 kg per shift.

3. Both routes count the same physical yarn; they can disagree only if an input is wrong — a mis-measured stitch length, a WPC/CPC taken in a different relaxation state, or a count not what its label says. A 10% gap is therefore a measurement alarm, not a rounding matter: re-measure l first, since every term passes through it.
